

Manifest Ten Parcels SOP

One episode manifests ten parcels. All ten are already sitting on their spots in the infeed lane when recording starts. Each parcel is moved onto the scale, its barcode is scanned to a good read, its weight band is read off the scale display and the band card, a carrier label is taken from the matching slot of the printer mock and pressed square onto the parcel's top face, and the parcel is lowered into the carrier bin carrying that same letter. The infeed lane is bare and all ten parcels lie labeled in the three bins when the episode ends.

The parcels are worked one at a time in printed spot order, 1 to 10. A parcel is finished and in its bin before the next one is lifted, and no two parcels are ever off their spots at once.

For each parcel the order never changes: move, scan, weigh, label, sort. No parcel is scanned before it is resting on the scale, no band is read while a gripper is still touching the parcel, no label is taken before the parcel has read, and no parcel leaves the platform before its label is pressed.

The right gripper does the primary work: it takes the scanner, reads the band, takes and applies the carrier label, and in Config L and M carries the parcel to its bin. The left gripper supports: in Config L and M it moves each parcel from the infeed lane onto the scale, and in every config it steadies the parcel for the label press. In Config R the feed and the sort change hands, as set out below. Neither gripper takes the other's zones, nothing is turned in the air, and a gripper opens only over the spot the thing in it is going to.

The desk is set up in one of three ways. Only the infeed lane moves; the scale, the scanner rest, and the printer mock are in the same place in all three. Where the lane takes a fixture's place, that fixture moves to the other side of the desk.

- **Config L:** the infeed lane is on the left side of the desk.
- **Config M:** the infeed lane is at the back-center, behind the scale; the band card stands just left of the scale.
- **Config R:** the infeed lane is at the back-right; the three carrier bins stand along the left side.

Where a step depends on the setup it says so on an **IF** line — look at the desk and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the infeed lane is on the left side (**Config L**), at the back-center (**Config M**), or at the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the infeed lane's side moves each parcel onto the scale — the **left gripper** in Config L and M, the **right gripper** in Config R. The gripper on the bins'

side carries each parcel to its bin — the **right gripper** in Config L and M, the **left gripper** in Config R. No arm reaches across the desk.

- **Fixed roles:** in every config the **right gripper** scans, reads the band, takes, applies, and presses the label, and the **left gripper** anchors the parcel's left wall for the press. The scale, the scanner rest, and the printer mock never move.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the infeed lane in the start zone for this episode's config, the scale at the center with the band card standing behind it (Config M: just left of it), the scanner rest at the front center, the printer mock at the front right, and all three carrier bins along the right side (Config R: along the left side).
3. The top face of the parcel on the scale is in frame from above, with the patch outline and all four edges of a label visible on it.
4. The near face of the parcel on the scale is in frame, so the barcode and the scanner head are both visible during a read.
5. The scanner's indicator lamp is in frame whenever the scanner is in the gripper.
6. The scale's display is in frame and its digits are readable, with nothing standing on the platform covering it, and the band card is in frame with all three lettered rows and their ranges readable.
7. Both arms are at home with grippers open.
8. The desk is clear of anything but the zones listed below.
9. The left gripper reaches the scale platform, (Config L and M) all ten infeed spots, and (Config R) all three carrier bins along the left side without the arm leaning out or reaching a joint limit.
10. The right gripper reaches the scale platform, the near face of a parcel on the platform, the scanner rest, all three printer mock slots, (Config L and M) all three carrier bins, and (Config R) all ten infeed spots at the back-right without the arm leaning out or reaching a joint limit.

Materials checklist

1. Ten **parcels** already resting flat and square on the ten printed **spots** of the infeed lane, one per spot, each with its barcode face toward the front edge. Nothing else stands on the desk except the zones below. The lane lies in the start zone for this episode's config:
 - **Config L:** the left side of the desk
 - **Config M:** the back-center, behind the scale, where the band card normally stands
 - **Config R:** the back-right, where bins B and C normally stand
2. Each parcel is a small rigid box with a flat top and a flat base, light enough and narrow enough for one gripper to lift from above by closing across its left and right walls.
3. Each parcel carries a printed **barcode** low on its near face, clear of the top edge.
4. Each parcel carries a printed **patch** outline in the middle of its top face, a little larger than a label, with its edges in line with the parcel's edges. Every patch is bare.
5. The ten parcels look alike from the outside and are weighted so that each one settles well inside one lettered range on the band card, never on or near the boundary between two ranges. The band cannot be told from the outside of the parcel.
6. The mix of bands across the ten parcels varies between episodes, and at least two bands are in use in every episode.
7. One digital **scale** sits at the center of the desk, switched on and zeroed, with its platform bare and its **display** reading zero. The display faces the front edge, the scale is set to one unit that does not change between episodes, and its auto power off is disabled or set long enough to outlast a whole episode.
8. One **band card** stands just behind the scale (Config M: just left of the scale), facing the front edge, printed with three lettered rows, **A**, **B**, and **C**, from lightest to heaviest, each row carrying the weight range that band covers. The ranges meet with no gap and no overlap and cover every parcel weight in use. The card stands outside both arms' zones and is never touched during an episode.
9. One **scanner** lying on the scanner rest, handle toward the front edge and head away. It is switched on and set to read continuously, so no trigger is pulled, and a good read shows one green flash of the indicator lamp with one beep.
10. One **printer mock** at the front right with three **outfeed slots** in a row, labeled **A**, **B**, and **C** left to right. Each slot holds a stack of ten **carrier labels** on a liner, printed face up, each printed with that slot's letter. The top label in each slot has its near edge already lifted free of the liner, so a gripper takes the lifted edge and the label comes away on a straight lift.
11. Three **carrier bins** standing in a row along the right side of the desk (Config R: along the left side), front to back: **bin A** nearest the front edge, then **bin B**, then **bin C**. Each is shallow, open topped, carries its letter on its near face, and is empty.
12. Every bin is deep enough and wide enough to hold all ten parcels lying flat if the bands run that way.

Workspace layout

- **Infeed lane:** the start zone — the left side of the desk (**Config L**), the back-center behind the scale (**Config M**), or the back-right (**Config R**) — ten printed spots in two rows of five. Spots 1 to 5 are the near row, left to right, and spots 6 to 10 are the far row, left to right. Working in printed order empties the near row first, so the feeding gripper never travels over a parcel still on its spot.
- **Scale:** the center of the desk. Every scan, band read, and label happens with the parcel resting on its platform.
- **Band card:** standing just behind the scale (Config M: just left of the scale), facing the front edge. It is read, never touched, and never leaned on.
- **Scanner rest:** the front center, just right of the scale. The scanner lies here before and after every read.
- **Printer mock:** the front right, three outfeed slots labeled A, B, and C.
- **Carrier bins:** the right side (Config R: the left side), in a row front to back, bin A nearest the front edge, then bin B, then bin C.

The left arm's zones are the left wall of the parcel on the scale, (Config L and M) the infeed lane, and (Config R) the three bins. The right arm's zones are the scale platform, the scanner rest, the printer mock, (Config L and M) the three bins, and (Config R) the infeed lane. Nothing carried crosses the midline except the short move from the infeed lane onto the scale.

Vocabulary

- **Arm assignment:** the right gripper scans, reads the band, takes and applies the carrier label, and in Config L and M carries the parcel to its bin. The left gripper steadies the parcel for the label press and in Config L and M moves each parcel from the infeed lane onto the scale. In Config R the right gripper feeds the scale from the back-right and the left gripper carries the parcel to its bin on the left. Neither gripper takes the other's jobs.
- **Start zone:** where the infeed lane lies at the start of the episode — the left side (**Config L**), the back-center (**Config M**), or the back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Bin side:** the side of the desk the three carrier bins stand on — the right in Config L and M, the left in Config R. The gripper on that side sorts.
- **Parcel:** one small box from the infeed lane. Its **near face** is the face toward the front edge and carries the barcode. Its **top face** carries the patch.
- **Spot:** one printed outline in the infeed lane, numbered 1 to 10. The spot number sets the working order and nothing else.
- **Barcode:** the printed code on the parcel's near face. It is scanned, never covered, and never touched by a gripper.

- **Good read:** one green flash of the indicator lamp with one beep, taken while the head is held still and square to the barcode. Nothing else counts as a read.
- **Scanner:** the tool on the scanner rest. Its **handle** has a flat grip section and its **head** carries the read window and the indicator lamp. A gripper closes on the handle and never on the head.
- **Platform:** the flat top of the scale. A parcel rests on it, centered, with its base flat and no part of it hanging over an edge.
- **Settled reading:** the number on the display holding steady for 2 seconds while the parcel sits alone on the platform with no gripper touching it. A number still climbing, falling, or flickering is not a reading.
- **Band:** the letter, A, B, or C, whose range on the band card contains the settled reading. The band decides both the label slot and the bin.
- **Carrier label:** one label from a printer mock slot, called the **label** below. It is the only thing applied to a parcel, and it carries the letter of the slot it came from.
- **Lifted edge:** the near edge of the top label in a slot, already free of the liner. It is the only part of a label a gripper ever touches.
- **Patch:** the printed outline on the parcel's top face that the label is laid inside.
- **Square:** the label lying inside the patch with each of its edges in line with the matching edge of the patch, no corner overhanging the outline, and no corner lifted.
- **Apply:** the right gripper lowering a label straight down into the patch with its adhesive face down and releasing the lifted edge once the label is lying on the top face.
- **Press:** the closed right gripper set down flat on the applied label, held still for 2 seconds, and lifted straight off. A press never slides and never rubs.
- **Anchor:** the closed left gripper resting against the outside of the parcel's left wall, level with the middle of the wall, holding the parcel still while the right gripper presses the label. An anchor never travels, never sits on the top face, never pushes the parcel across the platform, and lifts clear only after the right gripper is off the parcel.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower on:** carrying the thing level to its spot, lowering it straight down until it is resting, and releasing once it is resting. Nothing is dropped from height.
- **Settled:** the parcel or the scanner stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or tipping.
- **Release point:** on the scale platform, inside the patch, in a carrier bin, or on the scanner rest. A gripper opens nowhere else.

Steps

Steps 1 to 5 are one parcel. Run them on the parcel on spot 1, then spot 2, and so on through spot 10, finishing each parcel and its bin before the next parcel is lifted. Step 6 ends the episode once all ten are sorted.

Steps 1 and 5 depend on the config: in Config L and M the **left gripper** feeds each parcel onto the scale and the **right gripper** sorts it to a bin on the right; in Config R the **right gripper** feeds from the back-right and the **left gripper** sorts it to a bin on the left. Every other line in Steps 1 to 6 is the same in all three configs.

Step 1: Move the parcel onto the scale

Goal: the parcel resting alone and centered on the platform, barcode face toward the front edge.

Look where the infeed lane is before reaching for the first parcel.

- **IF the infeed lane is on the left side (Config L):** the **left gripper** closes across the left and right walls of the parcel on the **lowest numbered spot still occupied**, lifts it straight up, and carries it level to the right, to the **scale**.
- **IF the infeed lane is at the back-center (Config M):** the **left gripper** closes across the left and right walls of the parcel on the **lowest numbered spot still occupied**, lifts it straight up, and carries it level straight forward to the **scale**.
- **IF the infeed lane is at the back-right (Config R):** the **right gripper** closes across the left and right walls of the parcel on the **lowest numbered spot still occupied**, lifts it straight up, and carries it level forward and to the left, to the **scale**.

Then, in all three:

- Lower the parcel onto the middle of the **platform** until it is resting, and release.
- The feeding gripper lifts clear of the parcel and away from the scale.
- The other gripper stays clear of the scale for the whole step.

Take one parcel per trip. The parcel goes down the way it was staged, so nothing is turned on the way over.

Check: one parcel rests flat and centered on the platform, settled, its barcode face toward the front edge, its patch bare and readable from above, no part of it hanging over a platform edge, no gripper touching it, and the spot it came from is bare. If it lands off center or over an edge, lift it clear and lower it again.

Step 2: Scan the barcode to a good read

Goal: a good read on the parcel's barcode, with the scanner back on its rest.

- The **right gripper** comes down onto the flat of the **handle** on the scanner rest, closes across it, and lifts the scanner straight up.
- Bring the **head** square to the **barcode** on the parcel's near face, close to it but not touching it, and hold it still until the lamp flashes green and the scanner beeps once.
- Lift the scanner straight back, carry it to the **scanner rest**, lower it flat with the handle toward the front edge, and release.
- The right gripper lifts clear.

The tool moves, the parcel does not. If the lamp does not flash, draw the head back, bring it square to the barcode again, and hold for a second read. Never touch the parcel with the scanner and never move the parcel to bring its barcode to the head.

Check: the lamp flashed green and the scanner beeped once while the head was on the barcode, the scanner lies flat on its rest with the handle toward the front edge, and the parcel sits exactly where it was placed with its facing unchanged. If three attempts give no read, the barcode is defective: stop, log it as a system issue, and discard the episode.

Step 3: Read the weight band and take the matching label

Goal: one label from the slot matching the parcel's band, held by its lifted edge.

- With the parcel alone on the platform and **no gripper touching it**, wait for the **settled reading**, then find the lettered row on the **band card** whose range contains that number.
- The **right gripper** closes on the **lifted edge** of the top label in the **printer mock slot** **carrying that same letter**, lifts it straight up off the liner, and holds it level with the adhesive face down.

Read the band before you go to the printer mock, never the other way round. The band comes from the display and the card, and is never guessed from the size or the feel of the parcel.

Check: the display had settled and its number fell inside one lettered range when it was read, no gripper was on the parcel or the scale at that moment, and the label now in the right gripper carries that same letter. If the wrong slot was worked, lay that label back on its liner, leave it, and take one from the correct slot instead.

Step 4: Apply the label square on the top face

Goal: the label lying square inside the patch on the parcel's top face, pressed flat.

- The **left gripper**, closed, **anchors** the parcel on its **left wall** and holds it there for the whole step.
- The **right gripper** carries the label level over the parcel and lowers it straight down into the **patch**, with its edges in line with the patch's edges, until it is lying on the top face. Release the lifted edge.

- The **right gripper**, closed, presses down flat on the label, holds for 2 seconds, and lifts straight off.
- The **left gripper** lifts clear once the right gripper is off the parcel.

Line the label up over the patch before you lower it. Adhesive sticks once, so a label is placed by lowering it, never by sliding it into line after it has touched the parcel.

Check: the label lies inside the patch, square, with no corner overhanging the outline, no corner lifted, no fold, and no bubble; the letter on the label is readable from above and matches the band that was read; the barcode on the near face is uncovered; and the parcel is still centered on the platform. If the label landed off the patch, crooked, or folded on itself, peel it off, scrap it, and take a fresh label from the same slot.

Step 5: Sort the parcel to its carrier bin

Goal: the parcel lying flat in the bin carrying the same letter as its label, with the platform bare and the display back at zero.

- **IF Config L or M:** the **right gripper** closes across the left and right walls of the parcel, clear of the top face and the label, and lifts it straight up off the platform. **IF Config R:** the **left gripper** takes that same hold and lifts the parcel the same way, since the bins are on the left.
- Carry it level to the **carrier bin carrying the letter printed on its label**, lower it into the bin until it is resting flat on the bin floor or flat on the parcels already in it, and release.
- The sorting gripper lifts clear of the bin.

The letter on the applied label picks the bin. Lower the parcel all the way in rather than opening the gripper over the rim.

Check: the parcel lies flat in the bin whose letter matches its label, label face up, settled, fully inside the bin with no part riding the rim, no parcel already in the bin knocked on its side, the platform is bare, and the display has returned to zero. If the parcel went into the wrong bin, lift it straight out and lower it into the correct one.

Go back to Step 1 for the next parcel while any spot in the infeed lane is still occupied. When spot 10 is bare, go on to Step 6.

Step 6: Confirm the sorted parcels and end the episode

- Confirm all ten infeed spots are bare, ten parcels lie flat in the bins with each one's label square in its patch and matching the bin's letter, the scale platform is bare with its display at zero, and the scanner lies flat on its rest with the handle toward the front edge.
- Return both arms home, clear of the scale and the bins, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the ten parcels out of the bins, peel the label off each top face, and discard the used labels along with any that were scrapped. Labels are single use.
2. Confirm every patch outline is still readable and every barcode is still clean and unmarked.
3. Lay the infeed lane in the start zone for the next episode's config — the left side (Config L), the back-center (Config M), or the back-right (Config R) — with the band card just left of the scale in Config M and the three bins along the left side in Config R. Set the ten parcels back on the infeed spots, one per spot, flat and square, barcode face toward the front edge and patch bare.
4. Change which parcels carry which weights between episodes rather than running the same mix every time. Keep at least two bands in use and keep every parcel settling well inside one lettered range.
5. Restock any printer mock slot that has run down, printed face up on its liner, and lift the near edge of the new top label free of the liner in all three slots.
6. Check the scale: switched on, zeroed, platform bare, display reading zero, readable from the camera, set to the usual unit, and not timing itself out. Check the band card: standing behind the scale, facing the front edge, upright, and all three rows and ranges still readable.
7. Check the scanner: switched on, set to read continuously, and giving one green flash and one beep on a good barcode. Lay it flat on the scanner rest, handle toward the front edge.
8. Replace anything damaged: a parcel that is crushed or will not sit square, a barcode that is torn, smudged, or unreadable, a label that has lost its tack, or a bin that will not stand square.
9. Wipe the desk and confirm the infeed lane, the scale, the band card, the scanner rest, the printer mock, and the three bins have not shifted and that every printed outline and letter is readable.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config L (the infeed lane on the left side, the bins on the right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the desk — the infeed lane is not in the start zone for the config, or the band card or the bins are not where that config puts them; a gripper reaches across the desk for a parcel or for a bin; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper feeds the scale in Config L and M and the right gripper in Config R; the right gripper sorts to the bins in Config L and M and the left gripper in Config R; no arm reaches across the desk; the IF line followed is the one for the config on the desk).
- **Coaching note:** look where the infeed lane and the bins are before the first reach, then follow that config's IF lines through Steps 1 and 5.

Violation: Wrong order

- **Visible cue:** a parcel is scanned before it is resting on the platform, the band is read before the scan, a label is taken before the parcel has read, the parcel leaves the platform before its label is pressed, or it goes into a bin before the label is on.
- **SOP rule broken:** Steps 1 to 5, the order never changes: move, scan, weigh, label, sort.
- **Coaching note:** move, scan, weigh, label, sort. Finish the step you are in before you start the next one.

Violation: Parcels worked out of spot order

- **Visible cue:** a parcel is lifted while a lower numbered spot is still occupied, so the far row starts before the near row is empty and the left gripper travels over a parcel still on its spot.
- **SOP rule broken:** Step 1, take the parcel on the lowest numbered spot still occupied.
- **Coaching note:** count up from one. The near row clears before you reach across it.

Violation: More than one parcel off its spot at once

- **Visible cue:** a parcel is lifted out of the infeed lane while another is still on the platform or still in a gripper, two parcels are carried in one trip, or a parcel is parked on the desk beside the scale.

- **SOP rule broken:** Steps 1 to 5, finish each parcel and its bin before the next parcel is lifted.
- **Coaching note:** one parcel, one lap, all the way to its bin.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the scanner, takes a label, or carries a parcel to a bin; or the right gripper lifts a parcel out of the infeed lane.
- **SOP rule broken:** Steps 1 to 5, the left gripper moves the parcel onto the scale and anchors it, and the right gripper scans, labels, and sorts.
- **Coaching note:** left feeds and steadies. Right scans, labels, and sorts.

Violation: Moved on without a good read

- **Visible cue:** the scanner is returned and a label is taken with no green flash and no beep, or the head is waved past the barcode instead of being held still and square to it.
- **SOP rule broken:** Step 2, only one green flash with one beep counts as a read, and Step 3 is never started on a parcel that has not read.
- **Coaching note:** hold it still and wait for the flash. No flash, no label.

Violation: Scanner mishandled

- **Visible cue:** a gripper closes on the head or across the scanner at an angle instead of on the flat of the handle; the head touches the parcel, the barcode, the platform, or the desk; or the scanner is left on the parcel, in a bin, or still in the gripper when the label is taken, or dropped onto its rest rather than lowered.
- **SOP rule broken:** Step 2, the flat of the handle is the only part gripped, the head touches nothing, and the scanner is back flat on its rest before the label is taken.
- **Coaching note:** handle only, hover the head, then put the tool back.

Violation: Parcel moved to meet the scanner

- **Visible cue:** the parcel is slid, turned, tipped, or lifted off the platform during the scan, or a gripper holds the parcel up to the scanner head.
- **SOP rule broken:** Step 2, the tool moves and the parcel stays where it was placed with its facing unchanged.
- **Coaching note:** bring the head to the barcode. The parcel stays on the scale.

Violation: Band read while the parcel was touched or the display had not settled

- **Visible cue:** a gripper is on the parcel, on the platform, or on the scale body when the band is read, another parcel or the scanner is resting on the platform, or the label is taken while the number on the display is still changing.
- **SOP rule broken:** Step 3, the band is read with the parcel alone on the platform, no gripper touching it, and the display settled.

- **Coaching note:** hands off, let the number settle, then read the card.

Violation: Label taken from the wrong slot

- **Visible cue:** the letter on the label taken does not match the band card row holding the settled reading, or a slot is worked before the display has been read at all.
- **SOP rule broken:** Step 3, take the label from the printer mock slot carrying the same letter as the band.
- **Coaching note:** read the display, find its row on the card, then find that letter on the printer. Never the other way round.

Violation: Label taken by the wrong part

- **Visible cue:** a gripper pinches a label anywhere but its lifted edge, touches the adhesive face, or drags a label off its liner instead of lifting the edge straight up.
- **SOP rule broken:** Step 3, the lifted edge is the only part of a label a gripper touches.
- **Coaching note:** lifted edge only. A label that has touched the gripper face is already ruined.

Violation: Label not square

- **Visible cue:** the label lands outside the patch, overhangs the outline at a corner, sits at an angle to the patch edges, hangs over the edge of the parcel, or covers the barcode.
- **SOP rule broken:** Step 4, the label lies inside the patch with its edges in line with the patch's edges and no corner overhanging.
- **Coaching note:** line the label up over the patch before you lower it. The landing decides it.

Violation: Label peeled up and stuck down again

- **Visible cue:** an applied label is lifted off and pressed back down, slid into line after contact, or repositioned by peeling a corner.
- **SOP rule broken:** Step 4, a badly applied label is peeled off, scrapped, and replaced with a fresh label from the same slot.
- **Coaching note:** adhesive sticks once. Scrap it and take a fresh one.

Violation: Press skipped or rubbed

- **Visible cue:** the episode moves on from an applied label with no press, presses for less than the hold, dabs at it in several places, or slides the gripper across the label instead of holding it still.
- **SOP rule broken:** Step 4, one flat press held for 2 seconds and lifted straight off, never slid and never rubbed.
- **Coaching note:** down flat, count two, straight up.

Violation: Pressed without an anchor, or anchor in the wrong place

- **Visible cue:** the label is applied and pressed with the left gripper off the parcel, the anchor is taken only after the press has started, it lets go before the right gripper is clear, or the anchoring gripper sits on the top face, on the label, or pushes the parcel across the platform.
- **SOP rule broken:** Step 4, the left gripper anchors the outside of the left wall first, holds it still for the whole step, and lifts clear only after the right gripper is off the parcel.
- **Coaching note:** anchor the side wall first, let go last, and do not shove.

Violation: Parcel sorted to the wrong bin

- **Visible cue:** a parcel is released into a bin whose letter does not match the letter on its applied label, or it is released into a bin before the label has been read.
- **SOP rule broken:** Step 5, the letter printed on the applied label picks the bin.
- **Coaching note:** read the label on the parcel in your gripper, then find that bin.

Violation: Parcel gripped on the top face or on the label

- **Visible cue:** a gripper closes on, presses on, or drags across a parcel's top face, its patch, or its applied label during the anchor or the carry to the bin.
- **SOP rule broken:** Steps 4 and 5, grippers close across the left and right walls only and never on the top face or the label.
- **Coaching note:** side walls only. The top face belongs to the label.

Violation: Parcel dropped into the bin

- **Visible cue:** a gripper opens over a bin rim and the parcel falls the rest of the way, the parcel is tossed in, or it lands on its side, on end, or riding the rim and is left there.
- **SOP rule broken:** Step 5, lower the parcel into the bin until it is resting flat and release once it is resting.
- **Coaching note:** take it all the way in before you open the gripper.

Violation: Turned in the air

- **Visible cue:** a parcel or a label is rotated or flipped between the lift and the set down, so it lands facing a different way than it was carried.
- **SOP rule broken:** Steps 1, 3, 4, and 5, nothing is turned in the air; the way a thing was staged is the way it lands.
- **Coaching note:** the wrist does not have the range to turn it in transit.

Violation: Dragged instead of lifted

- **Visible cue:** a parcel or a label slides along the desk, the platform, or the liner still touching it, with no daylight under it in transit.

- **SOP rule broken:** Steps 1, 3, and 5, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on a parcel, a label, or the scanner without setting it down first.
- **SOP rule broken:** Steps 1 to 5, set a held thing down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the infeed lane, over the printer mock, over the scale body, over a bin rim, or over open desk, and what it held lands anywhere but where it was going.
- **SOP rule broken:** Steps 1 to 5, a gripper opens on the platform, inside the patch, in a bin, or on the scanner rest, and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a parcel is nudged, patted, or slid on the platform or in a bin after it is released, or a label is slid after it has touched the parcel, instead of the retry the step prescribes.
- **SOP rule broken:** Steps 1, 4, and 5, lift a misplaced parcel clear and lower it again, and scrap a misplaced label.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a parcel, a label, or the scanner falls to the desk or the floor short of its spot; the scale, the band card, the printer mock, the scanner rest, or a bin is struck and shifts; a parcel still on its spot is dislodged; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a parcel still on a spot, a parcel still on the platform, a parcel unlabeled or in a bin whose letter does not match its label, the scanner off its rest, or an arm away from home.
- **SOP rule broken:** Step 6, confirm the bare lane, the ten sorted parcels, the bare platform, and the scanner on its rest; return both arms home; then stop recording.

- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Scanner fault:** the lamp does not light, the scanner loses power, it beeps with no code in front of it, or it will not read a clean barcode held still and square after three attempts. Replace the scanner before the next episode.
- **Defective barcode:** torn, smudged, peeling off the parcel, or printed too faint to read.
- **Scale fault:** the display flickers between values, drifts while the parcel sits still, does not return to zero after a parcel is lifted off, switches itself off during the episode, or is unreadable from the camera.
- **A parcel's settled reading falls on the boundary between two ranges**, so no single band can be read. Restage that parcel's weight before the next episode.
- **Defective carrier label:** lost tack, torn on the lift, cut to the wrong size for the patch, or a near edge that will not stay lifted off the liner. Swap it out before the next episode.
- **Printer mock slot empty** when a band calls for it, or a stack that lifts off the liner as a block instead of one label at a time.
- **Defective parcel:** crushed, will not sit square on a spot or on the platform, too heavy or too wide for one gripper, or its top face patch is missing, misprinted, or already carrying a label.
- **A bin will not stand square, tips under a correct lower, or is too small** to take the parcels its band sends it.
- **The infeed lane, the scale, the band card, the scanner rest, the printer mock, or a bin shifts out of place** during the episode.

Annotation subtasks (from SOP)

1. Move one parcel from its infeed spot onto the scale
2. Take the scanner from its rest
3. Scan the parcel barcode to a good read
4. Return the scanner to its rest
5. Read the weight band from the scale display and the band card
6. Take one carrier label from the matching printer mock slot
7. Apply the label into the patch and press it
8. Carry the parcel to its carrier bin and lower it in

9. Return both arms home and end the episode